

# Alexander Cai

🌐 adzcaai | 🌐 adzc.ai | ✉ adzcaai@g.harvard.edu

## EDUCATION

---

**Harvard University**  
Ph.D. Computer Science

Cambridge, MA  
Sep 2025–Current

- Advised by Prof. Kianté Brantley
- Awarded Harvard Kempner Institute Graduate Fellowship

**Harvard University**  
S.M. Applied Mathematics  
A.B. Statistics and Computer Science (cum laude), Secondary in German Studies

Cambridge, MA  
Sep 2021–May 2025  
(concurrent)

- **Relevant coursework:** Foundations of Deep Learning, Topics in High-Dimensional Statistics, Neural Computation, Computing at Scale, Physical Mathematics, Operating Systems, Distributed Systems

## EXPERIENCE

---

**Brantley Lab (Harvard University)**  
*Graduate Research Assistant*

Cambridge, MA  
Sep 2025–Current

- Design imitation learning algorithms that generalize from few samples

**Harvard University**  
*Head Teaching Fellow, CS 1840: Introduction to Reinforcement Learning*

Cambridge, MA  
Sep 2026–Current

- Lead course staff for Prof. Kianté Brantley’s course: TF/CA hiring, course compute environment, exams, and an Embedded EthiCS module on RLHF

**Gershman Lab (Harvard University)**  
*Research Intern*

Cambridge, MA  
Jan–Aug 2025

- Implement a multitask RL algorithm based on the “multitask preplay” model from cognitive science
- Run distributed RL experiments with Weights & Biases on a high-performance compute cluster

**Harvard University**  
*Course Assistant*

Cambridge, MA  
Sep 2022–Dec 2024

- Led sections, designed course materials, and graded assignments for CS/Stat 184(0): Reinforcement Learning, CS 181(0): Machine Learning, AM 226: Theory of Neural Computation
- Authored course textbook for CS/Stat 184(0) also used for Cornell’s CS 4789

**QuantCo**  
*Software Engineering Intern*

Berlin, Germany  
Jun 2024–Aug 2024

- Designed a scalable Python framework for feature engineering

**Pehlevan Group (Harvard University)**  
*Research Intern*

Cambridge, MA  
May 2023–Aug 2023

- Implemented a novel unsupervised algorithm for 2D-to-3D video tracking based on transport operators
- Awarded Program for Research in Science and Engineering (PRISE) Fellowship

## DataStax

*Software Development Intern*

Santa Clara, CA, USA

Jun–Aug 2021, 2022

- Accelerated terabyte-scale bulk import tasks by up to 5x in DataStax’s Cloud Native Database
- Developed the `sstable-to-arrow` utility for reading Apache Cassandra database files into the fast Apache Arrow memory format

## PUBLICATIONS

---

- D. Rooryck\*, **A. Cai**\*, Yonatan Belinkov, David Alvarez-Melis, Kianté Brantley. “Feeding BabyLMs Macaroni: Code-Switching Curricula Cause Cross-Lingual Convergence,” 2026. EMNLP 2026 BabyLM Workshop. <https://arxiv.org/abs/2609.30535>
- J. Roggeveen\*, E. Wang\* et al. (incl **A. Cai**). “HARDMath2: A Benchmark for Applied Mathematics Built by Students as Part of a Graduate Class,” 2025. NeurIPS 2025. <https://arxiv.org/abs/2505.11774>
- **A. Cai**. “An Introduction to Reinforcement Learning,” 2025. Undergraduate thesis. <https://rlbook.adzc.ai>
- L. Tang, T. Shlomi, **A. Cai**. “Learning the Wrong Lessons: Inserting Trojans During Knowledge Distillation,” 2023. ICLR 2023 BANDS Workshop. <https://arxiv.org/abs/2303.05593>
- L. Tang, **A. Cai**, S. Li, J. Wang. “The Naughtyformer: A Transformer Understands Offensive Humor,” 2022. AAAI-23 Student Abstract. <https://arxiv.org/abs/2211.14369>
- W. Luo, R. Chin, **A. Cai**, G. Lin, B. Su, A. Zhang. “A tardiness-augmented approximation scheme for rejection-allowed multiprocessor rescheduling,” 2022. Journal of Combinatorial Optimization 44, 690–722. <https://link.springer.com/article/10.1007/s10878-022-00857-y>
- **A. Cai**. “Analyzing Cassandra Data Using GPUs,” 2021. <https://developer.nvidia.com/blog/analyzing-cassandra-data-using-gpus-part-1/>

## TALKS

---

- Kempner Lunch and Learn. Cross-lingual transfer in multilingual models. 8 October 2026.
- Kempner Lunch and Learn. Maximum entropy reinforcement learning. 16 September 2025.
- Kempner Lunch and Learn. A brief history of perspectives in language acquisition. 16 July 2025.

## POSTERS

---

- Kempner Institute Workshop: Learning Dynamics in Natural and Artificial Intelligence. Investigating code-switched curriculum learning in data-constrained multilingual modelling. 2 September 2026.
- Kempner Institute Spring Into Science. Grammar acquisition trajectories of developmentally plausible bilingual models. 21 May 2026.

## ACTIVITIES AND LEADERSHIP

---

### LGBTQ@GSAS

*Vice President*

Cambridge, MA

Sep 2026–Current

- Harvard Griffin GSAS student group

### Derek Bok Center for Teaching and Learning

*Teaching Development Fellow*

Cambridge, MA

Jun–Sep 2026

- Train undergraduate course assistants on teaching responsibilities and pedagogy

**InTouch (Harvard SEAS)***Peer Staffer*

Cambridge, MA

May 2026–Current

- Organize community building for InTouch, a peer-to-peer support network for SEAS graduate students

**Currier House, Harvard College***Non-Resident Tutor*

Cambridge, MA

Sep 2025–Current

- Mentor undergraduates and support residential community life as an affiliated graduate tutor

**AddisCoder***Teaching Assistant*

Addis Ababa, Ethiopia

Aug 2025, Aug 2026

- Teach computer science fundamentals to 120 high school students

**The Harvard Crimson***Director of Graphics*

Cambridge, MA

Mar 2024–Dec 2024

- Designed interactive visualizations to illustrate student newspaper articles
- Directed team of 10 undergraduate designers and programmers
- Finalist for the Society of Professional Journalists Mark of Excellence Award

**Harvard Undergraduate Machine Intelligence Community (HUMIC)***Co-Director of Fellowships*

Cambridge, MA

Apr 2022–Dec 2023

- Design and teach fellowship program about modern ML and AI to 70 undergraduates
- Lead weekly paper readings on state-of-the-art developments in ML and AI

**Harvard College Project for Asian and International Relations (HPAIR)***Co-Director of Technology*

Cambridge, MA

Oct 2021–Aug 2023

- Designed and built application portal at my.hpair.org, enabling over 20,000 applications
- Integrate Firebase with BigQuery and Google Sheets to provide a transparent and accessible view

## SKILLS AND INTERESTS

---

**Programming:** Jax RL ecosystem, PyTorch, C++, Apache Arrow, TypeScript, Firebase, Redis, SQL**Languages:** English (native), Mandarin (HSK 6), German (B2), French (B2)**Awards:** Germanic Languages and Literatures Book Prize, CyberTitan III National Silver Medalist, University of Toronto National Book Award, Canadian Open Mathematics Challenge Gold Award, Alberta Winter Games Men's Sabre Gold Medalist**Interests:** Reinforcement learning, linguistics, web development, historical fiction, trombone, musical theater